

Article

Variable Properties of Auditory Scene Analysis in Music

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Abstract: This article explores the variable properties of auditory image analysis during the perception of musical works, which are influenced by the specific elements to which the listener directs their attention. Traditional analyses of musical compositions typically involve brief comparisons with auditory phenomena described in scientific studies, such as those by A.S. Bregman. However, these analyses are often limited, offering only a narrow perspective on the works. In contrast, the approach presented in this article extends the theories and experiments developed by Bregman and others, providing a more comprehensive understanding of entire compositions or selected sections rather than focusing solely on isolated passages. This broader framework enhances auditory image analysis and serves as a foundation for further research. The expanded analysis integrates within music theory, enabling a deeper exploration of musical structures, particularly in the context of perceiving multilayered music where multiple sound sources may share similar acoustic features. The author illustrates how acoustic and perceptual factors contribute to complex mental representations through graphic and musical examples. To substantiate the claims, classical works by composers such as F. Chopin, A. Guilmant, and J.S. Bach are analysed, further highlighting the variable properties of auditory image analysis.

Keywords: multivariantism; perception; listening; auditory scene analysis; music; music theory



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1. Introduction

Understanding how the human mind processes and organises sound stimuli is central to analysing music perception, both for musicians and persons without musical backgrounds. Traditional psychoacoustic research, conducted in a laboratory setting, focuses on analysing perceptual streams in which isolated sounds merge into simple sequences. Still, experiments of this kind are not directly concerned with analysing musical works. Although such investigations are useful for acousticians and psychologists, they are often confined to a fragmentary view of music, which yields simplified interpretations and limits the applicability of the findings, especially for those with musical training.

The perception of an auditory scene is a complex process that plays a key role in perceiving and interpreting music. The analysis presented here introduces a broader context to better understand elaborate musical structures, especially in multilayered pieces where numerous sound sources share similar acoustic characteristics. Exploring the variable properties in auditory image analysis offers more profound insights into music perception as it highlights its multivariant nature.

Music processing in the mind is strongly informed by subjective and cultural factors which influence one's interpretation of different elements in a piece of music. For this reason, perceptions of music tend to vary considerably, depending on the social context, personal musical experiences, and the emotional associations it evokes.

2. Methodological and Theoretical Challenges of Research on Auditory Scene Analysis in Light of the Scientific Literature

There is an appreciable gap between the analysis of single sounds, which is the focus of contemporary psychoacoustic experiments, and the difficulty involved in analysing an auditory scene composed of multiple musical sequences. One of the numerous factors affecting the perceptual organisation of music, including the segregation of auditory streams, is timbre. However, given the variability of elements within timbre itself, systematising the entire phenomenon and arranging it into a clear framework is a challenge. Music poses a challenge to perception, being one of the most unpredictable acoustic environments a human may encounter, whereby the analysis of the auditory scene requires the brain to separate and organise multiple overlapping sounds, often emitted simultaneously by different sources, such as the instruments in an orchestra or the voices in a choir. Music becomes particularly problematic for perception when potential sound sources are numerous while the sounds possess similar characteristics (apart from timbre), such as rhythm, pitch, harmony, etc. (Adamo 2011; Hubbard 2010). In such circumstances, the number and the characteristics of sound sources are difficult for the brain to determine. In addition, music often uses some form of auditory illusion, which greatly hampers analysis (Deutsch 1974, 1975, 1983, 1992; Tougas and Bregman 1990). Due to structural complexity and the use of perceptual ambiguities, musical pieces present a great challenge to the brain's ability to recognise and analyse sound scenes, often producing a sense of multilayered and equivocal sound perception in the listener (Pressnitzer et al. 2011).

The listener's perception of two distinct sound scenes depends on which perceptual element they focus their attention on. This phenomenon is often discussed in the literature based on the results of psychoacoustic studies. The sensation accompanying the perceiver may be compared to switching between two simultaneously existing images, for example, background–figure and vice versa (Bigand et al. 2000). During a looped playback of a sequence or a series of repeated sounds, the listener may experience a threshold point at which sounds integrate into one stream (due to perceived similarities) or segregate into two perceptual streams (due to perceived differences) (McAdams 2019). Depending on the acoustic conditions, such as the changing tempo or frequency separation, the listener may perceive the same sounds in various ways. Perceivers may “switch” between different perceptual modalities but are unable to perceive several interpretations simultaneously. The brain opts for one of the possible perceptual organisations while other interpretations temporarily remain “in the background” (McAdams and Bregman 1979). Thus, it may be inferred that the integration or segregation of musical material into one or two perceptual streams as part of auditory scene analysis—already at this stage—relies on the principle of the bistability of the perceived stimuli, whereby the listener chooses one of the two emerging figures (Bregman 1990, 2004; Winkler et al. 2009; Lakatos et al. 2013; Riecke et al. 2015; Lerousseau et al. 2021).

The characteristic vocal structuring in polyphonic music enables the listener to perceive multiple melodies or instrumental lines, where their brain has to decide which sounds belong to the same melody and which belong to other layers of sound (Rasch 1978; Hausfeld et al. 2021). Participants in the experiments have been capable of identifying melodies in pieces containing two or three melodic lines at the same time with a high degree of accuracy, which suggests that the perception of more than two melodies is possible (Gregory 1990). There may also be moments when perception oscillates between two (or more) possible interpretations. Consequently, the perceiver may perceive all sounds as one integrated stream and separate sound streams at other times (Sauvé 2017; Disbergen et al. 2018; Disbergen 2020). The analysis of melody, harmony, and bass line bears on the identification of the piece, while the resulting combination of various

mental representations that the listener arrives at facilitates their realisation that they are confronted with distinct renditions of the same work (different arrangement versions). When perceiving the entire piece—in its mixed state—the perceiver has to disentangle it mentally to find differences in, for example, a given rendition ([Salamon et al. 2012](#)).

Often enough, composers deliberately create musical structures that cause the listener's brain to switch between different layers of sound. Through pseudo-polyphony, several distinct voices emerge from a single melodic line in certain pieces, leading to varied perceptual interpretations in the listener. Such manipulation of perceptual mechanisms may be employed by composers with a view to bring the listener closer to the boundary between the segregation and integration of perceptual streams ([Davis 2011](#)).

3. Bistability as a Foundation of Multivariability

Because of bistability, the listener can only perceive one of two possible images at any given time, depending on the perceptual element that draws their attention at a particular moment. In this theory, the choice of one of two interpretations may derive from a number of factors, such as expectations, knowledge of the world, attentional focus, and previous experience ([Raftopoulos 2011](#); [Leleu-Merviel 2014](#)). Bistability is a well-documented phenomenon in perceptual psychology, often studied using optical illusions such as Rubin's vase, the "rabbit–duck" figure, or the Necker cube ([Brandimonte and Gerbino 1993](#); [Moreno-Sanchez et al. 2016](#); [Filevich et al. 2017](#)).

Bistability is approached as the outcome of a dynamic interaction between the properties of the stimulus properties and human states of mind, where the perceptual change can be modulated both by sensory and cognitive inputs. This is because the brain cannot reconcile two interpretations simultaneously, resulting in the latter competing ([Takei and Nishida 2010](#)). Bistability is subject to factors such as the symmetry of the shape and the context in which the figure is perceived ([Verstijnen and Wagemans 2004](#); [Macpherson 2006](#)). The perception of bistable images can be determined by the interaction of external stimuli (properties of the image) and internal stimuli (related to various states of mind and cognitive processes) ([Moreno-Sanchez et al. 2016](#)). Auditory bistability occurs when the perceiver hears one sound that may be interpreted twofold, leading to spontaneous changes in perception. For instance, in the case of auditory streams, a series of tones may be perceived as a single, continuous stream or as two separate perceptual streams ([Kondo et al. 2012](#)). Perceptual bistability requires the flexibility of switching between different perceptual representations, which may enhance problem-solving abilities and creativity. Mechanisms that modulate bistable perception may, therefore, be useful in developing creative thinking through perceptual transformation processes ([Baron 2020](#); [Rodríguez-Martínez 2023](#)).

Bistability usually involves simpler figures perceived through vision or hearing ([Wilson 2012](#)), whereas multivariability (also known as multivariance or multistability) denotes the analysis of more complex structures, such as advanced forms of music or graphics. In consequence, multivariability represents a significant extension of bistability, presuming the recognition of more than two images. In music, this may include the capacity to perceive pieces in various ways, depending on the timbre, rhythm, harmony, or compositional structure. With graphic works, multivariant engagement reveals distinct layers or dimensions that coexist in a single work, offering the viewer a richer aesthetic experience. This capacity of human perception to discern numerous forms of perceptual heterogeneity underlies the development of theories of perception and aesthetics.

While bistable perception, such as figure/ground segregation, is well documented in visual studies, its application to auditory perception requires careful consideration. First, it should not be assumed that auditory figure/ground segregation operates identically to its

visual counterpart. Auditory perception encompasses dimensions distinct from those of visual perception.

Listeners often rely on various auditory grouping mechanisms that facilitate the perception of distinct auditory streams within complex musical textures. These mechanisms include, for example, the following:

- Similarity, which involves grouping elements based on shared characteristics (e.g., similar loudness or identical attack times of sounds within chords that form harmonic structures) (Pinna et al. 2016).
- Proximity, which refers to grouping elements based on spatial or temporal closeness (e.g., temporal adjacency of sounds or proximity in pitch) (Barlach 2021).
- Common Fate, which entails grouping elements according to shared dynamic characteristics (e.g., simultaneous temporal shifts in sounds or the transposition of entire sonic structures) (Chen and Fan 2006; Spillmann 2006).
- Closure, which involves the mental completion of missing elements within auditory structures (e.g., the perception of sounds that are acoustically masked by other tones or noise) (Mather 2006; Cabral et al. 2020; Lidwell et al. 2010).
- Good Continuation, which refers to the integration of overlapping structures into a coherent whole (e.g., the blending of sounds into a unified auditory background), also understood as the perception of a given auditory figure despite it being masked by other concurrent sounds (e.g., auditory masking effects in music) (Ockelford 2004; Jackendoff and Lerdahl 2006).

These mechanisms enable listeners to switch between different interpretations of the same musical material, creating a dynamic and flexible auditory scene.

It is crucial to distinguish between the roles of attentional mechanisms and perceptual organisation. Rubin demonstrated that figure/ground segregation and multistability do not inherently depend on attentional focus (Rubin 1915). However, perceptual pop-out effects—where certain sounds stand out due to their distinct features—do interact with attentional selection. This interaction necessitates a nuanced understanding of how bottom-up sensory information and top-down cognitive processes shape auditory perception (Treisman and Gelade 1980).

Phenomenal transparency, a concept often discussed in visual perception, also plays a significant role in auditory scene analysis (Metzinger 2017). In music, this manifests when overlapping auditory streams allow for partial segregation without complete perceptual dominance. Listeners can perceive certain elements as both part of the background and the foreground, depending on how attention is directed and how grouping cues are integrated.

4. The Emergence of Multivariant Structures: The Example of Geometric Figures and Sound Sequences

Figure 1 shows a field which, during analysis, may yield distinct mental figures for the perceiver, depending on the pattern on which they focus their gaze. The field consists of triangles in two lighter and two darker shades of grey and vertexless, rounded “squares” in the same shades of grey. The further use of the word “square” in the description will mean noticing or presenting a figure whose appearance only resembles the field of a square. The graphic materials in Figures 1–9 are all ambiguous images, as the viewer, focusing attention on particular elements that make up the image, will identify a different main figure or figures.

Figure 2 shows the emergence of six identical figures in the viewer’s mental image, each composed of eight triangles. The lighter-toned triangles are located at the bottom and top of each figure, while the darker triangles are to be found on the left and right.

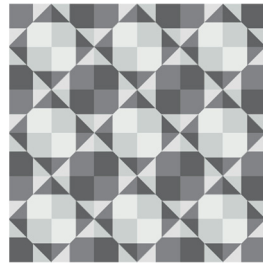


Figure 1. The field shown in Figure 1 may be interpreted by the perceiver in different ways, as shown in Figures 2–9.

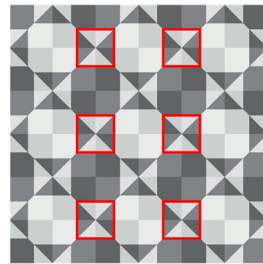


Figure 2. The illustration presents one of many figures that can be interpreted by the perceiver.

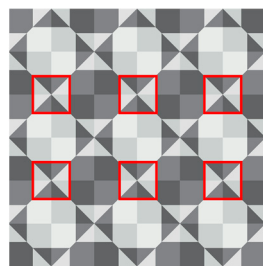


Figure 3. The illustration presents one of many figures that can be interpreted by the perceiver.

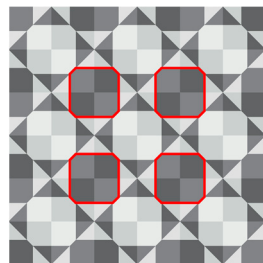


Figure 4. The illustration presents one of many figures that can be interpreted by the perceiver.

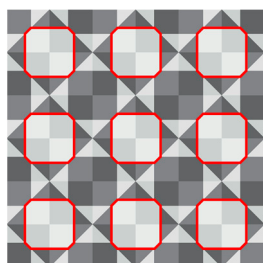


Figure 5. The illustration presents one of many figures that can be interpreted by the perceiver.

An analogous—though further reversed—situation may be observed in Figure 3. Again, the perceiver sees six identical figures, but they are aligned differently than in Figure 2. Also, the colour scheme in each identified image is inverted, as darker shades of grey are at the top and bottom of each image, while lighter shades are on the left and right.

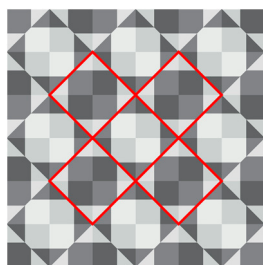


Figure 6. The illustration presents one of many figures that can be interpreted by the perceiver.

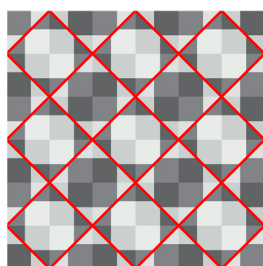


Figure 7. The illustration presents one of many figures that can be interpreted by the perceiver.

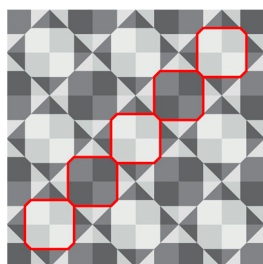


Figure 8. The illustration presents one of many figures that can be interpreted by the perceiver.

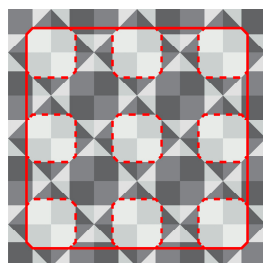


Figure 9. The illustration presents one of many figures that can be interpreted by the perceiver.

The visual analysis of the field in Figure 4 indicates that the viewer may combine the darker shades into a single figure, resulting in four dark squares in identical shades of grey.

In Figure 5, nine identical figures may be observed: squares composed of lighter shades of grey, in contrast to Figure 4. Interestingly, combining lighter colours has resulted in more figures and a different arrangement than in Figure 4.

The figure obtained in Figure 6 comprises the following:

- Four squares with sharp vertices (the central square is omitted), where each of the images is a square rotated at a 45° angle.
- Five squares with sharp vertices (the central square is noticed), where each of the images is a square rotated at a 45° angle.
- Nine squares with sharp vertices (the central squares are omitted), where each of the images is a square rotated at a 45° angle.
- Thirteen squares with sharp vertices (the central squares are omitted), where each of the images is a square rotated at a 45° angle. Interestingly, when the perceiver

distinguishes the thirteen figures, four of them have their colours reversed relative to the remaining nine shapes identified at the same time.

- In Figure 8, five squares with rounded corners form a kind of diagonal. The elements that make up each square are alternately lighter or darker. This case illustrates only one of the phenomena that may occur because it would also be possible for the perceiver to see, e.g.,
 - squares with rounded corners that mimic the second diagonal (still alternately lighter or darker);
 - squares with one lighter colour, which run vertically, dividing the field into two halves;
 - squares with one lighter colour which run horizontally and also divide the field into two halves;
 - other figure variants.¹

Finally, Figure 9 demonstrates that it is possible to analyse an image as a large square with rounded corners (red solid line) whose field exceeds the previously described figures. In addition, the perceiver may arrive at a mental image of further smaller structures inside that large field, such as the minor squares with rounded corners composed of lighter elements marked by the dotted line.

Figures 10–15 show a two-bar fragment of Ludwig van Beethoven's *Piano sonata C Major* (Waldstein), op. 53, no. 21, *Allegro con brio*, m. 14–15, a piece that may be perceived and interpreted differently by listeners depending on the auditory element which draws their primary attention. The example presented here is intended to demonstrate the varied analytical properties of the auditory image which arise in the course of exposure to musical works, directly transposing earlier examples in the form of computer graphics (Figures 1–9) into music.

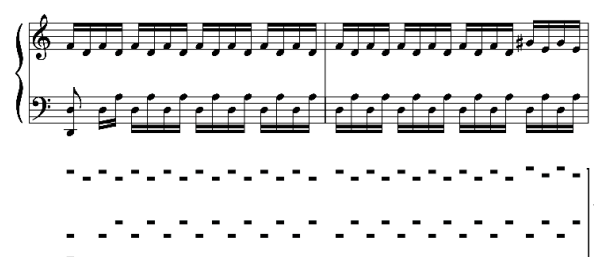


Figure 10. Ludwig van Beethoven, *Piano sonata C Major* (Waldstein), op. 53, no. 21, *Allegro con brio*, m. 14–15; mental representation for variant no. 1.

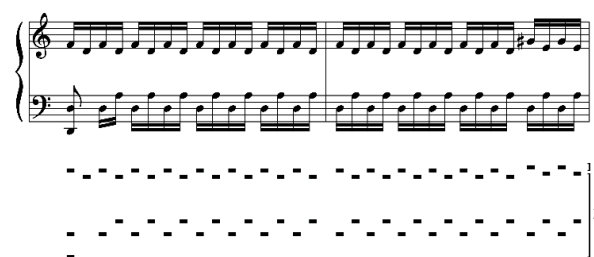


Figure 11. Ludwig van Beethoven, *Piano sonata C Major* (Waldstein), op. 53, no. 21, *Allegro con brio*, m. 14–15; mental representation for variant no. 2.

Figure 10 illustrates variant no. 1. The tones are integrated by the perceiver into a singular, continuous perceptual stream. This method of auditory perception is typical for individuals without musical training, who cannot analytically distinguish and mentally organise distinct pitches from the complex array of sounds they encounter. Additionally, fatigued listeners, even those with musical training, might overlook important perceptual details during the listening process. This can lead to interpreting sounds in

the manner described by this perceptual variety as their ability to concentrate on auditory stimuli diminishes.

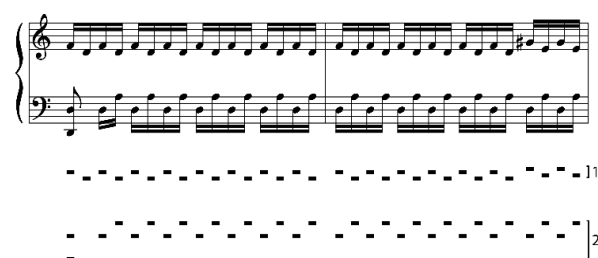


Figure 12. Ludwig van Beethoven, *Piano sonata C Major (Waldstein)*, op. 53, no. 21, *Allegro con brio*, m. 14–15; mental representation for variant no. 3.

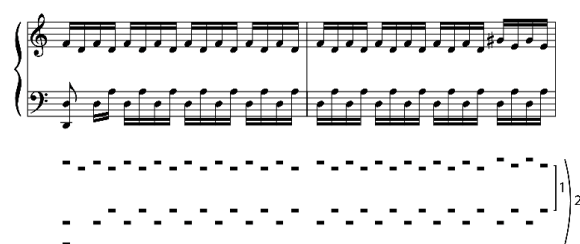


Figure 13. Ludwig van Beethoven, *Piano sonata C Major (Waldstein)*, op. 53, no. 21, *Allegro con brio*, m. 14–15; mental representation for variant no. 4.

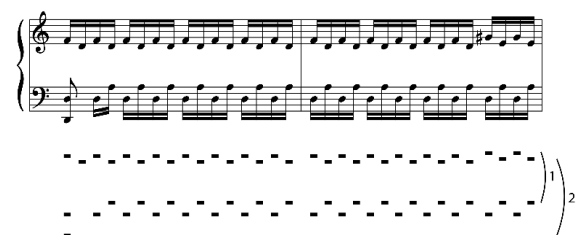


Figure 14. Ludwig van Beethoven, *Piano sonata C Major (Waldstein)*, op. 53, no. 21, *Allegro con brio*, m. 14–15; mental representation for variant no. 5.

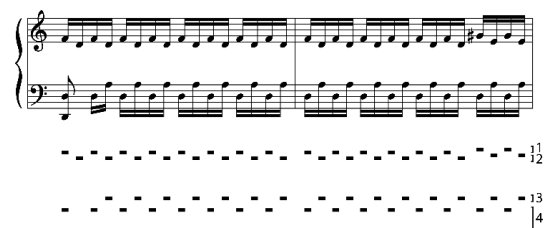


Figure 15. Ludwig van Beethoven, *Piano sonata C Major (Waldstein)*, op. 53, no. 21, *Allegro con brio*, m. 14–15; mental representation for variant no. 6.

Figure 11 illustrates the scenario where the recipient processing auditory stimuli amalgamates them into a configuration akin to variant no. 2. Under these circumstances, contingent upon the recipient's concentration, attentiveness, and the manner in which the composition is executed, the observer may construe alternate notes within the sequence of sixteenth notes as a singular perceptual continuum. The uppermost tones extracted from the sequence might serve as the foundation for the cognitive construction of the tune (the first stream), whereas the remaining tones could be attributed to the secondary stream, comprising the harmonic structure of the musical composition.

In Figure 12, a distinct third perceptual variation is depicted. The sequence of sixteenth notes, characterised by the higher pitches (performed by the musician's right hand), constitutes the initial unified perceptual stream. In contrast, the subsequent stream is formed

by the lower pitches (produced by the left hand). This mode of stimulus perception emerges when the observer partitions both sequences into analogous halves based on musical notation and the execution technique involving the left and right hands. For instrumentalists familiar with this composition and its prior renditions, performance nuances related to hand motor skills may suggest such a method of segregating sounds into two perceptual streams.

In Figure 13, it is possible to observe an intriguing fourth perceptual variation—the notes arranged vertically on the musical score (harmony sounds) give rise to two distinct perceptual streams. These perceived streams intertwine and are heard alternately, necessitating the listener's focused attention on individual sounds to maintain the continuity of perception. The first stream comprises sounds with a narrower interval between them, colloquially described as “inside” or “in the middle”. Conversely, the listener perceives the outermost sounds—the lowest and highest pitches—as constituting the secondary perceptual stream.

Figure 14 depicts a fascinating acoustic phenomenon, which can be perceived as perceptual variation no. 6. The observer consolidates sounds into perceptual streams in a cross-sectional manner. A lower pitch, played within a sixteenth-note sequence of the left-hand segment, is amalgamated into a perceptual stream with the subsequent lower pitch of another sixteenth-note sequence (not originating from the same harmonic vertical), as notated in the right-hand segment. Subsequently, a second perceptual stream is constructed following the aforementioned pattern, albeit in this instance, the stream is composed of cross-combined higher pitches. This manner of integrating sounds into distinct streams only arises during performances at high tempos, owing to the constraints of human perception. The rapid tempo renders the observer incapable of accurately and precisely discerning the pitch of each individual tone. This tempo-induced blurring of focus on a particular perceptual image (difficulty in recognising the precise arrangement of sounds) prompts the listener to amalgamate sounds based on alternative, not necessarily correct or fully comprehended, perceptual criteria.

In the next figure, the emergence of four distinct perceptual streams is observable. The sixth variation presents a formidable challenge to perceive, as the listener can only attend to two simultaneous streams. However, by adjusting and shifting attention to other concurrent sounds in a specific manner, the listener can discern the additional streams. This represents the approach employed by seasoned musicians, who cultivate specific auditory aptitudes and skills through their specialised musical training, enabling them to analyse sounds discerningly. In this scenario, the listener can discern up to four discrete streams, each potentially interpreted as one of four separate voices.

5. Equipment Utilised During the Listening Sessions

The listening setup consisted of the following components:

- Laptop: HP Pavilion Aero 13-be0802nw, running Microsoft Windows 11 operating system;
- Audio interface functioning as a sound card: Steinberg UR824;
- Wired headphones: Beyerdynamic DT 770Pro, with an impedance of 80 ohms.

The volume of the played music was established during the preliminary listening session using pilot tracks. The audio path gain was adjusted to provide the utmost comfort for listening to various musical pieces (Rosinski 2021). The peak loudness reached 95 phons during the loudest sections of the music, while the average loudness was approximately 73 phons. The sample rate and resolution settings in the audio interface during playback were 44.1 kHz and 16 bit.

6. Listening Method, Listening Sessions, and Audio Material

A key methodological aspect of this study was a comprehensive analysis of musical works, combining investigations of recordings and their scores. The recordings provided insights into the actual acoustic realisation of the pieces, encompassing timbre, dynamics, and expressivity, while the scores offered structural and formal interpretative frameworks that complemented the listening experience. This two-tiered approach revealed the multi-dimensional nature of musical perception, illustrating how identical acoustic stimuli can be interpreted differently depending on prior knowledge, cultural context, and situational factors such as mood or listening focus.

The author acknowledges that various performances of the same composition can lead to differing perceptual effects due to changes in tempo, dynamics, articulation, performers' interpretative nuances, or the acoustics of the recording space. However, the principal aim of this article is to demonstrate that identical acoustic stimuli, stripped of performance-related variables, can still evoke complex auditory interpretations. Future research will address the impact of differing performances on perception, enabling a more nuanced exploration of this intricate phenomenon. Additionally, the studies will include group analyses to examine how individual factors, such as musical education or genre preferences, shape perceptual variations. These investigations aim to complement the present findings, offering deeper insights into the interplay between performance context and subjective listening experiences.

This analysis draws on the author's extensive expertise in auditory stream analysis, both as a music theorist and a practitioner with advanced analytical skills in auditory perception. The decision not to replicate this experience with respondents stems from the author's prior psychoacoustic experiments conducted on groups of musicians and non-musicians, which yielded extensive data on how these groups integrate and segregate sounds. These findings have provided a clearer understanding of the perceptual mechanisms influencing the formation of mental figures. It was thus determined that further similar studies would not significantly contribute new insights, as the existing data are sufficient to formulate credible and detailed conclusions (Rosiński 2021, 2023, 2024a, 2024b, 2024c, 2024d).

The author deliberately assumes the role of a music theorist focusing on the study of sound perception in real-time listening contexts. Within this approach, particular emphasis is placed on the process of interpreting acoustic stimuli, seeking connections between sound structure and the mental figures that may emerge in the listener's mind during auditory experiences. This perspective captures the immediacy and ephemerality of musical experience, enabling an exploration of how listeners transform raw auditory material into complex mental constructs.

All musical works analysed in this study were personally reviewed by the author under strictly controlled acoustic conditions, minimising disturbances caused by playback quality or room acoustics. To ensure consistency, the material was played using headphones with a neutral sound profile in an acoustically isolated environment. This method eliminated potential interference, enhancing the reliability of observations and conclusions. The focus was placed on music with complex textures and rich sonic layers, which cannot be fully appreciated in laboratory settings that employ test tones exclusively.

The following works were selected for auditory scene analysis in the context of perceptual stream occurrence:

- Frédéric Chopin, *Etude C Major*, op. 10, no. 7, *Vivace*, m. 1–11; *Etude F Minor*, op. 10, no. 9, *Allegro molto agitato*, m. 1–16; *Etude A flat Major*, op. 10, no. 10, *Vivace assai*, m. 1–9, performed by: Janina Fialkowska, from the record: *Frédéric Chopin—Etudes, Sonatas & Impromptus*, 2554, record label: ATMA Classique, 1997;

- Alexandre Guilmant, *Organ sonata: Finale*, op. 42, no. 1, *Allegro assai*, m. 6–8, 93–95, 105–110, performed by: Olivier Vernet, from the record: *Alexandre Guilmant—Sonate no. 1, opus 42, Sonate no. 5, opus 80*, 03700, record label: Ligia Digital, 2013;
- Johann Sebastian Bach, *Sonata D Minor*, no. 3, BWV 527, *Vivace*, m. 17–36, performed by: Marie-Claire Alain, from the record: *Bach—Complete works for organ*, record label: Erato/Warner Classic, 2018, CD 1.

7. Author's Analysis—An Analysis of Sheet Music Confirming the Presence of Variable Auditory Perceptions

In Frédéric Chopin's piece, *Etude C Major*, op. 10, no. 7, *Vivace*, m. 1–11, shown in Figure 16, two ways of perceiving the heard sounds can be observed, and what is very important—the conscious listener can freely alter (direct) their perception to variant 1 or 2, depending on exactly which elements they focus their attention on:

- Variant no. 1: the highest notes performed with the right hand, treated as the melody by the listener, form the first perceptual stream, while the second stream consists of the remaining notes performed with hands, creating the harmonic background.
- Variant no. 2: the lowest sounds written in the part of the left hand are interpreted by the listener as the first perceptual stream, while all the other notes written in the parts of both hands are interpreted as the second perceptual stream—the harmonic background.



Figure 16. Frédéric Chopin, *Etude C Major*, op. 10, no. 7, *Vivace*, m. 1–11. Source: <https://imslp.org/wiki/Special:ImagefromIndex/168185/cy28> (accessed on 2 February 2024).

It is worth mentioning that the listener might completely overlook some of the sounds of the melody in their perceptions—especially when there is a very small interval between the sounds classified by the listener as harmony and/or melody (most often, the occurrence of a minor third or major second between sounds performed at the same time). A small pitch difference can result in the lack of segregation of higher sounds (van Noorden 1977, 1975), thus integrating them into harmony. The rather high tempo and the concentration of small subdivisions in the part of the right hand result in the possibility that the listener's attention may be unable to keep up with all the nuances and alterations in the rapidly changing auditory image (Brochard et al. 1999).

Figure 17 presents an excerpt from Frédéric Chopin's piece, *Etude F Minor*, op. 10, no. 9, *Allegro molto agitato*, m. 1–16. This section reveals several different possible ways of perceiving, which are worth mentioning:

- Variant no. 1: The listener hears two perceptual streams. One of them consists solely of sounds assigned to the right-hand part, while the other stream is formed by the sounds of the left hand performing sixteenth-note passages, which are interpreted as the harmony of the piece.
- Variant no. 2: The perceiver also hears two perceptual streams, although the first stream is formed by the sounds assigned to the right hand together with the two highest notes from each sequence consisting of six sixteenth notes. The highest notes performed with the left hand are played simultaneously with the melody in the right hand's part (their attack is identical), which causes the integration of both sounds into a single perceptual stream (Bregman 1990). In this case, the listener also perceives a melody created from dyads because at moments where the two highest notes performed in the part of the left hand occur, the difference in the pitch of the sounds written in the left- and right-hand parts decreases, causing the integration of sounds.
- Variant no. 3: In this variant, the listener is able to perceive the third perceptual stream, which is formed from Cb3 or C3 sounds in the left-hand part. A large amount of Cb3 and C3 sounds enables the listener's mind to identify a separate perceptual figure, the sounds of which integrate into another stream due to the multiple repetitions of the same sound in the rhythmic pattern (Bregman 1990). The remaining streams can be interpreted by the listener as specified in the descriptions of variants 1 and/or 2.



Figure 17. Frédéric Chopin, *Etude F Minor*, op. 10, no. 9, *Allegro molto agitato*, m. 1–16. Source: <https://imslp.org/wiki/Special:ImagefromIndex/168185/cy28> (accessed on 2 February 2024).

Another excerpt from a piece by Frédéric Chopin, shown in Figure 18, demonstrates that the composer used some elaborate arrangements of sounds in his works, which can be successfully analysed today in terms of the wide variety of perceptual insights, which also indicates how music is full of various contents that are still relevant, thanks to which we can continue exploring them all the time:

- Variant no. 1: The listener perceives only one perceptual stream. In this case, the rapid progression of sounds and the high tempo of the piece disrupt the accurate perception of all sounds, making the analysis incomplete and omitting important

elements that could cause the segregation of the streams (Dannenbring and Bregman 1976; Grimault et al. 2001).

- Variant no. 2: Two perceptual streams are observed, one consisting of sounds belonging to the part of the right hand and the other one being built of sounds belonging to the part of the left hand.
- Variant no. 3: The perceiver also perceives two perceptual streams. In this particular case, the sounds produced with both hands overlap, and the listener hears two alternating harmonic streams. All the dyads performed in the right-hand part combine with the sound in the bass line, which resounds at the same time—forming one of the streams. The second stream is formed as a result of the overlapping of single notes recorded in the right- and left-hand parts, which resound at the same time, thus being interpreted by the listener as separate from the rest.
- Variant no. 4: The listener is able to follow three streams. In this case, the dyads written in the part of the right hand form the first of the streams, while the second is built as an alternative melody formed from single notes, still performed in the part of the same hand. Similarly to variant no. 3, the listener perceives two alternating streams, which they can focus on interchangeably. The precise tracing of the streams is not possible because of the high tempo used by the composer combined with small subdivisions, making it difficult for the listener to focus their attention carefully (Repp 2009; Botte et al. 1997).
- Variant no. 5: It is a hybrid variant, which means that for certain individuals (listeners), there is a possibility of a fourth perceptual stream occurring as a result of the separation of one of the perceptual streams into two, both constructed based on sounds produced with the pianist's left hand; however, this is not an expected phenomenon.
- Variant no. 6: It is also possible to mix the different variants described earlier under the sixth variant, understood as a hybrid. This happens for a very simple reason: the individual analysing the incoming sounds can shift their attention in short moments to other components of the auditory image. Then, in short periods of time, they perceive different acoustic realities, but due to the high tempo, the overlapping sounds, and the differences in the pitch scales of various rapidly progressing sounds or consonances, the listener is unable to analyse the entire section according to one precisely executed principle (variant). Overlooking certain acoustic events or paying too much attention to others may distort the auditory image and result in anomalies in the perception of sound stimuli (Dykstra and Gutschalk 2013).



Figure 18. Frédéric Chopin, *Etude A flat Major*, op. 10, no. 10, *Vivace assai*, m. 1–9. Source: <https://imslp.org/wiki/Special:ImagefromIndex/168185/cy28> (accessed on 2 February 2024).

Figure 19 shows a three-measure-long excerpt from the *Organ Sonata: Finale*, op. 42, no. 1, *Allegro assai* by Alexandre Guilmant. The written sounds, through various classifications made by the listener, will be arranged into different mental figures:

- Variant no. 1: Sounds performed in the part of the right hand form the first perceptual stream, while the ones in the left hand form the second.
- Variant no. 2: Through the interval proximity of sounds written in a similar pitch scale, the interpretation of a given stream is not uniform. Depending on the focus of the listener's attention, the perceiver may direct their interpretation of the stimuli to melodic (left-hand part) and/or harmonic (right-hand part) elements. In this case, the listener may also perceive the melody and the harmony interchangeably, depending on how they interpret the incoming sounds.
- Variant no. 3: Each highest sound from the harmonic vertical played by the instrumentalist with the right hand is treated by the listener as the melody and assigned to the first stream. The second stream is formed by the remaining sounds from the triads (making them dyads), which are interpreted as belonging to the harmony. The third stream is built on the basis of the sounds performed with the instrumentalist's left hand.
- Variant no. 4: This variant is identical to variant no. 3 with one difference—the notes performed with the left hand split into two streams. One of them is formed from the repeated E4 notes (every other sound), while the other consists of all the remaining notes performed by the instrumentalist with the left hand.



Figure 19. Alexandre Guilmant, *Organ sonata: Finale*, op. 42, no. 1, *Allegro assai*, m. 6–8 and 93–95 (both fragments are identical). Source: <https://imslp.org/wiki/Special:ImagefromIndex/290298/cy28> (accessed on 20 February 2024).

Figure 20 presents another excerpt from the same movement of Alexandre Guilmant's piece and measures 105 to 110. Different perspectives on the incoming sounds may result in a wide range of mental representations:

- Variant no. 1: All sounds performed in the right-hand part are treated as belonging to the first stream. In this case, the listener hears a descending arpeggio. The sounds written in the part of the left hand form the second stream, and those in the pedal part form the third stream.
- Variant no. 2: The proximity of the pitch scales of sounds written in the left- and right-hand parts in measures 105 and 106 makes the sounds merge into one integral perceptual stream. Subsequent measures, as well as the sounds of the pedal, are interpreted according to the description of variant no. 1.
- Variant no. 3: The highest notes in the right-hand part (quarter notes) form the first melodic perceptual stream. The second stream is composed of the sixteenth notes written in the part of the right hand and all the notes in the part of the left hand. The third and final perceptual stream is formed from the notes performed in the pedal part.
- Variant no. 4: As in variant no. 3, the highest notes performed in the part of the right hand (quarter notes) form the first melodic perceptual stream. The second stream is built from the sixteenth notes written in the part of the right hand. Mainly

from measure 107 to 110, sounds performed in the part of the left hand, through a significantly increased pitch scale compared to the sounds written in the part of the right hand, are segregated and become the basis for the formation of the third stream. The sounds performed by the instrumentalist in the pedal form the last (fourth) perceptual stream.

- Variant no. 5: This variant can emerge due to the listeners using different perceptual segregation processes in individual measures. The structure of the mentioned stream is not homogeneous, and it can be based on the combination of all four previously described variants, as well as the creation of new sub-variants. For example, in measures 109 and 110, the quarter-note passage performed by the instrumentalist with his left hand can be integrated into a single perceptual stream with the pedal sounds since the applied rhythm is identical. The closeness of the tones to each other in the pitch scale causes the phenomenon of masking (because of the same attack time of those sounds) (Wang and Sogin 1990; Wang 2005).



Figure 20. Alexandre Guilmant, *Organ sonata: Finale*, op. 42, no. 1, *Allegro assai*, m. 105–110. Source: <https://imslp.org/wiki/Special:ImagefromIndex/290298/cy28> (accessed on 20 February 2024).

Figures 21 and 22 present an excerpt from a piece by Johann Sebastian Bach, which is very interesting in terms of the occurring perceptual streams, as it can be perceived by the listener in unusual ways:

- Variant no. 1: In measures from 17 to 24, in the case of different timbres being used in two manuals (left- and right-hand parts), there are two main perceptual streams, the dominant stream changing every second measure, as one stream mostly consists of notes written in measures 17, 19, 21, and 23, while the second stream consists of notes written mostly in measures 18, 20, 22, and 24. The third stream in this variant is formed by eighth note lines performed in the pedal part.
- Variant no. 2: This variant is almost identical to variant no. 1, although in this case, the listener omits the sounds of the eighth notes and the quarter notes extended by the first sixteenth note from a triplet written in the next measure (such rhythmic notation is found in both left- and right-hand parts in measures 17 to 24).
- Variant no. 3: In measures from 17 to 24, in the case of similar timbres being used in two manuals (the left- and right-hand parts), the sounds are cross-combined. The sixteenth-note triplets, written one time in the left and one time in the right-hand part, are integrated into the first stream. The second perceptual stream also consists of cross-combined sounds written in the parts of both hands, although the rhythmic pattern occurring here is an eighth note and a quarter note extended by the first sixteenth-note triplet written in the next measure. The third stream is formed by eighth-note lines performed in the pedal part.



Figure 21. Jan Sebastian Bach, *Sonata D Minor*, no. 3, BWV 527, *Vivace*, m. 17–30. Source: https://imslp.hk/files/imglnks/euimg/7/7b/IMSLP867601-PMLP152882-Bach_Triosonate_d-Moll.pdf (accessed on 10 March 2024).



Figure 22. Jan Sebastian Bach, *Sonata D Minor*, no. 3, BWV 527, *Vivace*, m. 31–36. Source: https://imslp.hk/files/imglnks/euimg/7/7b/IMSLP867601-PMLP152882-Bach_Triosonate_d-Moll.pdf (accessed on 10 March 2024).

In measures from 25 (Figure 21) to 36 (Figure 22), the listener can observe the transition of voices (transferring the continuity of the perceptual stream) between the left- and right-hand parts. Since the construction of the stream, in this case, is not homogeneous because the sixteenth-note triplets and the regular sixteenth-note passages draw the listener's attention much more, they can integrate into a single perceptual stream (based on rhythmic similarities) in the measures chosen by the listener.

In measures from 29 to 31, the simultaneous eighth-note lines written in the left hand and the pedal parts may be integrated by the listener into a single perceptual stream due to the identical rhythm and the same attack time.

In measure 35, all the parts may integrate into a single perceptual stream based on similarities in rhythm and attack time, although from measure 36, the pedal part begins to form a separate perceptual stream again due to the continuation of the rhythm from the previous measure, which will be clearly distinguishable from the background of the quarter notes and eighth-note pauses in the parts of both hands (rhythmic restraint), and so it will be noticed by the listener.

8. Discussion

The phenomenon of bistability is well documented in the literature on visual perception, particularly in classical optical illusions. The application of that theory to the analysis of musical sounds represents a novel approach, which nevertheless highlights the fact that auditory perception can be just as complex as visual perception. This article demonstrates that listeners may spontaneously switch between different interpretations of the same sound stream, leading to manifold auditory experiences. The extension of the bistability theory to the multivariability paradigm suggests that perception may involve multiple parallel interpretations, not just two alternatives, as in classical bistability.

In music, multivariability manifests itself in the listener's ability to perceive a composition in different configurations, depending on external factors such as tempo, dynamics, articulation, and internal conditions such as the degree of concentration, musical experience, and familiarity with the piece. Perception will be contingent on the element of the composition to which the listener directs their attention. For instance, the melody may be perceived as the dominant stream, while other elements, such as harmony, may be the background. Alternatively, both may be perceived as separate streams.

The results of the analyses show that the experience of the listener has a major impact on musical perception. Those conversant with music are capable of distinguishing more perceptual layers in musical pieces than individuals without formal training. Less-experienced listeners perceive music as a homogenous whole, while those with more expertise will separate perceptual streams that contain melody, harmony, rhythm, and other components. This phenomenon is not static, since the perception of music may change even in an experienced listener due to diminished attention or increased fatigue. Perception may shift during a single interpretation and in repeated exposure to the same piece, resulting in various auditory experiences. Traditional musical analysis often prioritises technical aspects such as harmony, formal structure, and instrumentation. However, from the standpoint of perception, music can be studied in a more dynamic fashion, taking into account how the listener actively shapes the perception of sounds.

Findings from this research suggest that music theory could benefit from incorporating the perceptual aspects of multivariability, which would likely facilitate a more complex analysis of musical works that goes beyond the musical notation to account for the variety of possible perceptual modalities. The results of the presented analyses could be potentially applied in practice by performers and sound engineers alike. An awareness of multivariability may encourage musicians to shape the dynamics, articulation, and tempo more deliberately to manipulate listeners' perceptions and evoke specific auditory impressions. Further studies may contribute to developing analytical tools that allow for the variability of perception given the context, such as the acoustic properties of a concert hall or sound reproduction technology.

Although these results are eloquent enough, one cannot fail to note certain limitations involved in analysing perceptual streams in music. The phenomenon of multivariability might not occur in equal measure in all listeners. Sound perception is subjectively variable and may prove difficult to put into a definite theoretical framework. In addition, individual differences in music perception, including auditory experience, emotional state, as well as acoustic and technical factors, may affect how an individual perceives music. Prospectively, it may be worthwhile to focus on investigating individual differences in music perception, e.g., among professional musicians, amateurs, or persons with hearing deficits, in order to better understand the perceptual mechanisms across a spectrum of audiences.

9. Conclusions

In this article, the author applies the theory of auditory scene analysis (ASA) to the study of music characterised by complex structures and multilayered textures. While previous laboratory-based research has often focused on simple sound sequences, such as isolated tones or short melodic-rhythmic motifs, extending ASA theory to the intricate and multifaceted art of musical composition necessitates methodological adaptation and the development of new analytical tools. These findings have significant implications for music education, underscoring the importance of training in auditory attention and fostering a conscious engagement with the diverse sonic layers present in music. A deeper understanding of perceptual processes not only enhances listeners' appreciation of music but also informs more deliberate approaches to musical creativity, performance, and pedagogy.

Auditory perception, characterised by its intricate and highly analytical nature, reveals the phenomenon of hearing in all its remarkable complexity. The analysis of auditory imagery illustrates how the perception of sound can vary significantly depending on the surrounding context. Understanding this phenomenon is especially critical for musicians, as it offers them a novel perspective on the auditory scene and its interpretation. The multiplicity of ways in which identical auditory stimuli can be perceived highlights an untapped potential within listeners that warrants thorough scientific investigation. Rather than viewing the formation of distinct mental representations in response to the same auditory stimulus as an erroneous interpretation, researchers should recognise it as a manifestation of listeners' unique perceptual abilities, which can emerge in particular auditory contexts.

The proposed method of auditory analysis underscores how much remains to be explored and understood in this domain. Researchers frequently investigate how we process various sound structures, with an emphasis on the fleeting, transient nature of auditory perception—a characteristic especially pronounced in music, as opposed to the more static experiences found in the visual arts. The insights listeners derive from musical pieces can be partially explained through an understanding of perceptual streaming, which, depending on the listener's focus, sequence of musical elements, and other factors (such as cognitive load or momentary attention), leads to the formation of diverse musical structures (Cf. [Bregman 1990](#); [Bregman and Ahad 1996](#)).

The existence of perceptual streams within music demonstrates that beyond the traditional boundaries of music theory, there are additional layers of complexity shaping how sound is received and interpreted. These perceptual dynamics remain insufficiently studied, leaving gaps in our understanding of the cognitive processes underlying musical perception. It is essential to investigate these dynamics further to better comprehend the mechanisms that guide the brain's interpretation of auditory stimuli. Importantly, this novel approach does not diminish previous achievements in the field but expands upon them. As with any evolving discipline, music theory benefits from incorporating new concepts such as perceptual streaming, which plays a crucial role in shaping listeners' experiences and influencing the interpretation of musical works.

To conclude the discussion, it is worth posing the question of whether the research findings would retain their validity if the analysed musical examples did not exhibit a "streaming" form. It seems pertinent to consider whether more "dispersed" sound structures, characterised by highly expansive configurations—as found in the works of composers such as Webern, Stockhausen, and Boulez—facilitate or, conversely, hinder the perception of complex layers of auditory streams. The musical examples presented in this article appear to promote stream-based perception, raising the question of how these processes might manifest in different musical contexts.

In this regard, the role of the tonal system, which dominates Western music, also warrants consideration in shaping the perception of auditory streams. Does tonality support the segregation of such streams, or, conversely, does it complicate perceptual processes, making it more challenging to discern multilayered sound structures? It seems that the analysis of streaming phenomena should encompass both a neurocognitive perspective—focusing on the perceptual mechanisms responsible for processing one or multiple auditory streams—and compositional strategies, in which the layering of textures represents a deliberate artistic choice. A particularly illustrative example in this context is the passage from Johann Sebastian Bach's Sonata in D Minor, No. 3, BWV 527, Vivace, encompassing measures 17–23. The melodic lines in the upper voices form a subtle, concealed texture, meticulously crafted and strategically integrated into the structure of the piece, highlighting the composer's mastery in manipulating complex perceptual relationships.

Delving into manipulating sound through detailed auditory scene analysis opens the door to discovering a range of psychoacoustic phenomena that affect listeners, regardless of their individual cognitive approach. Advancing research in this area holds the potential to significantly broaden the limits of auditory cognition and deepen our understanding of the human capacity for sound perception, particularly as it relates to the experience of music. By introducing these perceptual phenomena into the study of music theory, scholars can uncover new dimensions of musical works, offering fresh and previously overlooked perspectives for research.

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Note

- ¹ The author focuses exclusively on the most significant and readily discernible figures relevant to this study, acknowledging that individual listeners may interpret the emerging figures differently. Additional variants and their descriptions are deliberately omitted, as in a scientific context, such details are unnecessary and do not contribute to the advancement of knowledge. Subsequent figures are derived from the mechanisms and principles already outlined, rendering further elaboration redundant.

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